

Statistical Analysis of HR Workforce Data

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1 Question 1: Conduct a thorough exploratory data analysis of the HR dataset:

1.1 Before analysis

The HR dataset was explored to identify its structure, summarise the numeric variables, and detect patterns, outliers, and relationships before later inference. Based on the HR context, Salary was expected to be positively associated with YearsOfExperience and JobLevel, and to be more skewed than the other numeric variables.

1.2 Analysis

1.2.1 (a) Begin by loading the dataset into R. Report its dimensions, the types of variables present, and the overall structure.

```
# Load the HR dataset from the CSV file
hr <- read.csv("data/group5.csv")
```

```
# Report the dimensions of the dataset
dim(hr)
```

```
## [1] 172 18
```

```
# Examine the overall structure and variable types
str(hr)
```

```
## 'data.frame': 172 obs. of 18 variables:
## $ EmployeeID : chr "EMP_0001" "EMP_0002" "EMP_0003" "EMP_0004" ...
## $ Department : chr "Sales" "Operations" "Sales" "HR" ...
## $ Age : int 38 25 29 29 32 34 36 31 37 22 ...
## $ Gender : chr "Male" "Female" "Female" "Male" ...
## $ JobLevel : chr "Mid" "Mid" "Entry" "Mid" ...
## $ EducationLevel : chr "High School" "High School" "High School" "Bachelor" ...
## $ YearsOfExperience: int 16 1 6 5 9 9 12 7 14 1 ...
## $ Salary : int 94802 56709 56130 62830 99207 81296 70401 53093 89776 46449 ...
## $ PerformanceScore : int 3 2 1 5 1 5 5 3 3 4 ...
## $ SatisfactionScore: int 3 4 8 1 9 4 8 8 3 5 ...
## $ OvertimeHours : int 2 13 4 18 13 15 12 6 12 16 ...
## $ TrainingHours : int 92 91 84 1 41 7 0 8 35 93 ...
## $ ProjectsCompleted: int 7 15 3 13 19 9 14 17 13 19 ...
## $ TeamSize : int 10 12 8 10 11 6 5 11 11 7 ...
## $ RemoteWorkDays : int 1 3 2 2 3 0 4 1 4 1 ...
## $ AttendanceRate : num 0.906 0.976 0.95 0.988 0.89 0.995 0.989 0.91 0.936 0.921 ...
## $ PromotionCount : int 5 0 2 1 3 3 4 2 4 0 ...
## $ OfficeLocation : chr "Austin" "New York" "Chicago" "Chicago" ...
```

The CSV used for analysis contained 172 observations and 18 variables. The data included character variables such as EmployeeID, Department, Gender, JobLevel, EducationLevel, and OfficeLocation, and numeric variables such as Age, YearsOfExperience, Salary, PerformanceScore, SatisfactionScore, AttendanceRate, PromotionCount, and other workforce measures.

Overall, the dataset was structured appropriately for employee-level HR analysis.

- 1.2.2 (b) Compute and present summary statistics (mean, median, standard deviation, quartiles) for all numeric variables in the dataset. Use visualizations to detect any unusual observations or outliers.

```
# Select all numeric variables
numeric_vars <- hr[, sapply(hr, is.numeric)]

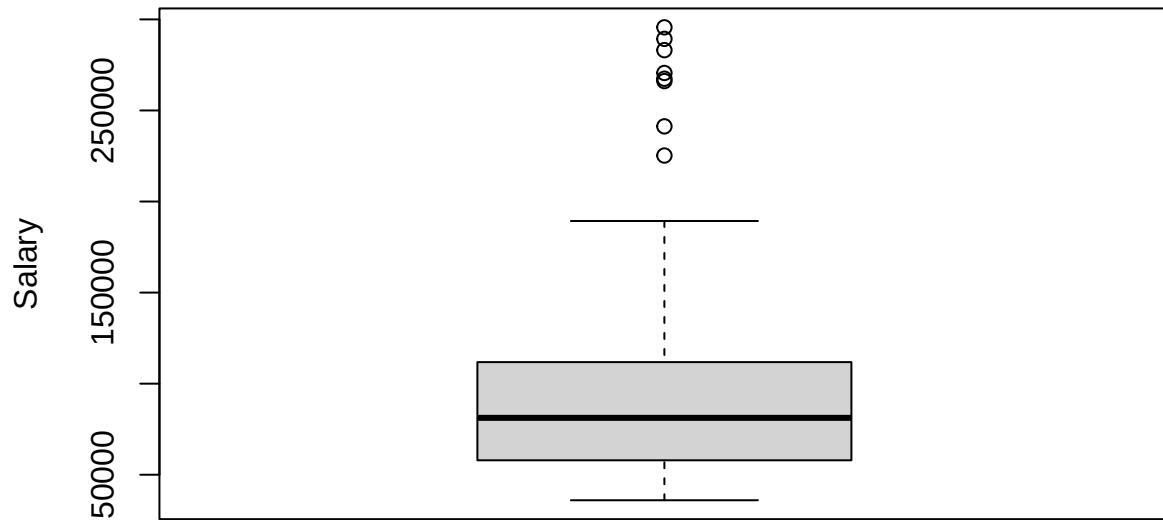
# Compute summary statistics for all numeric variables
summary_stats <- data.frame(
  Mean = sapply(numeric_vars, mean),
  Median = sapply(numeric_vars, median),
  SD = sapply(numeric_vars, sd),
  Q1 = sapply(numeric_vars, quantile, 0.25),
  Q3 = sapply(numeric_vars, quantile, 0.75)
)

summary_stats
```

##	Mean	Median	SD	Q1	Q3
## Age	33.7732558	33.000	7.72561509	28.000	39.00000
## YearsOfExperience	9.9534884	9.000	7.43927774	3.000	15.00000
## Salary	94362.0581395	81228.500	52568.11057509	58052.750	110659.50000
## PerformanceScore	3.0465116	3.000	1.41344403	2.000	4.00000
## SatisfactionScore	5.7267442	6.000	2.75952044	3.000	8.00000
## OvertimeHours	9.4186047	9.000	5.83489226	4.000	14.25000
## TrainingHours	47.9418605	47.500	30.28533108	24.000	75.25000
## ProjectsCompleted	10.7267442	11.000	5.64865920	5.750	16.00000
## TeamSize	8.1802326	8.000	3.39712020	5.000	11.00000
## RemoteWorkDays	2.3953488	2.000	1.72886786	1.000	4.00000
## AttendanceRate	0.9290814	0.928	0.04279127	0.891	0.96625
## PromotionCount	2.6220930	3.000	1.92617557	1.000	5.00000

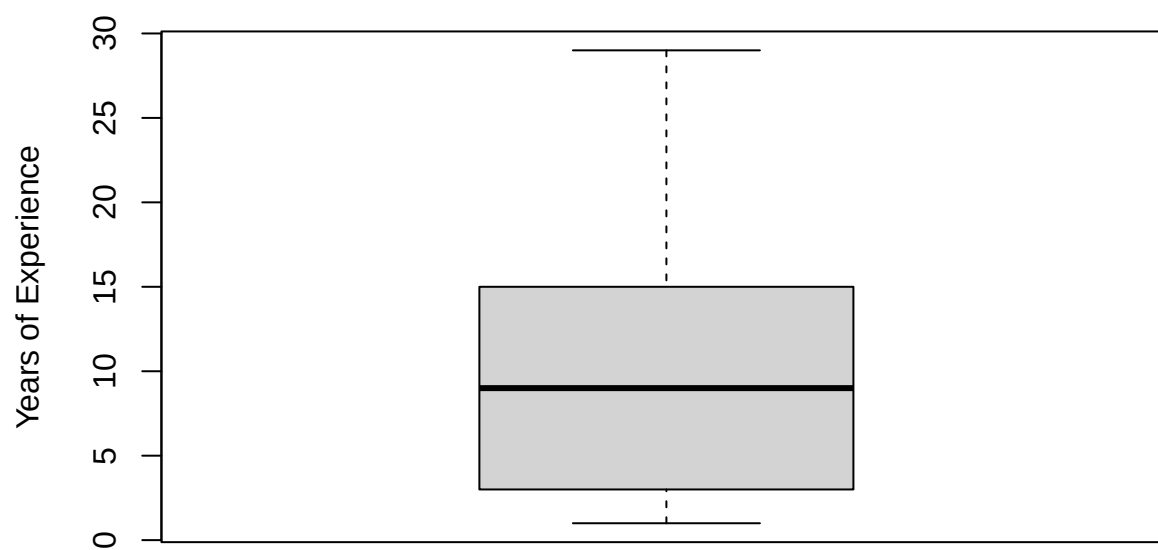
```
# Create boxplots for selected numeric variables to detect unusual observations or outliers
boxplot(hr$Salary, main = "Boxplot of Salary", ylab = "Salary")
```

Boxplot of Salary



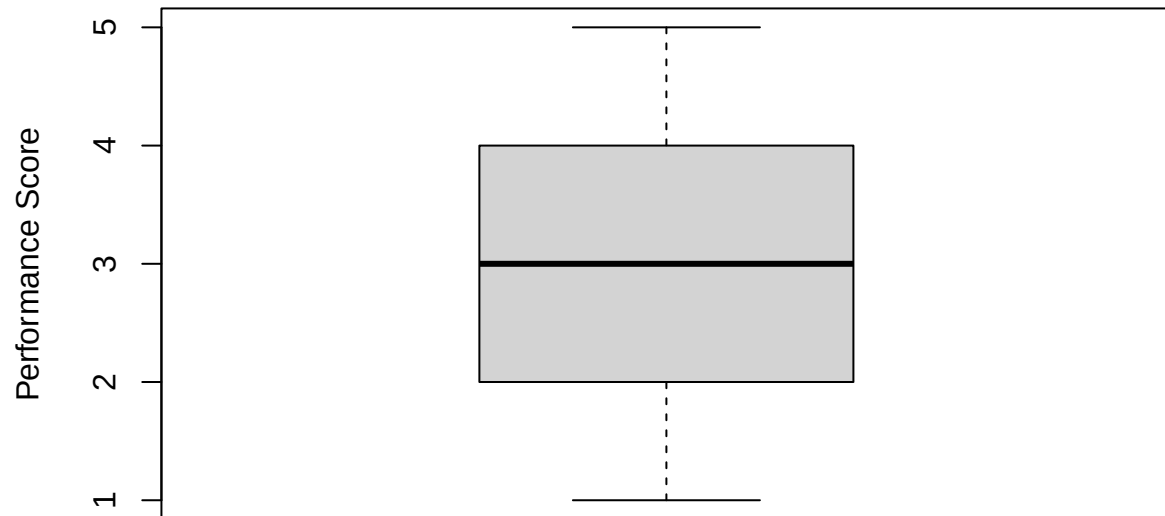
```
boxplot(hr$YearsOfExperience, main = "Boxplot of Years of Experience", ylab = "Years of Experience")
```

Boxplot of Years of Experience



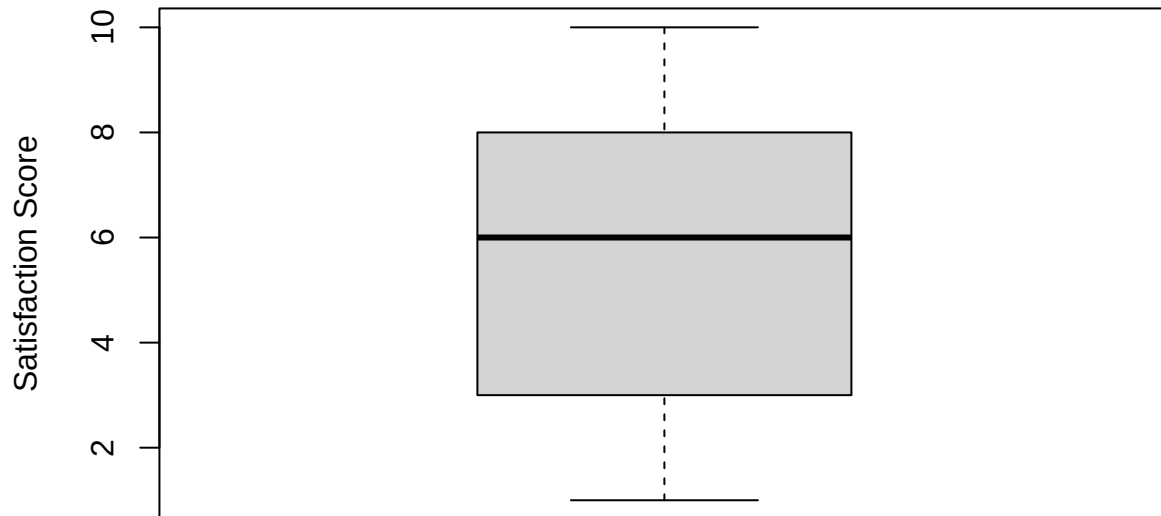
```
boxplot(hr$PerformanceScore, main = "Boxplot of Performance Score", ylab = "Performance Score")
```

Boxplot of Performance Score



```
boxplot(hr$SatisfactionScore, main = "Boxplot of Satisfaction Score", ylab = "Satisfaction Score")
```

Boxplot of Satisfaction Score



Summary statistics showed substantial variation across numeric variables. Salary was the most dispersed variable, with mean = 94,362.06, median = 81,228.50, and SD = 52,568.11. Because the mean was higher than the median, Salary appeared positively skewed. Its quartiles (Q1 = 58,052.75, Q3 = 110,659.50) also suggested wide spread.

YearsOfExperience had mean = 9.95 and median = 9, indicating a moderately experienced workforce. PerformanceScore was centred around 3, while SatisfactionScore was centred around 6. AttendanceRate showed low variation, with mean = 0.929 and SD = 0.043.

The boxplots showed that Salary had the clearest outliers, mostly at the upper end, likely reflecting higher-paid senior employees.

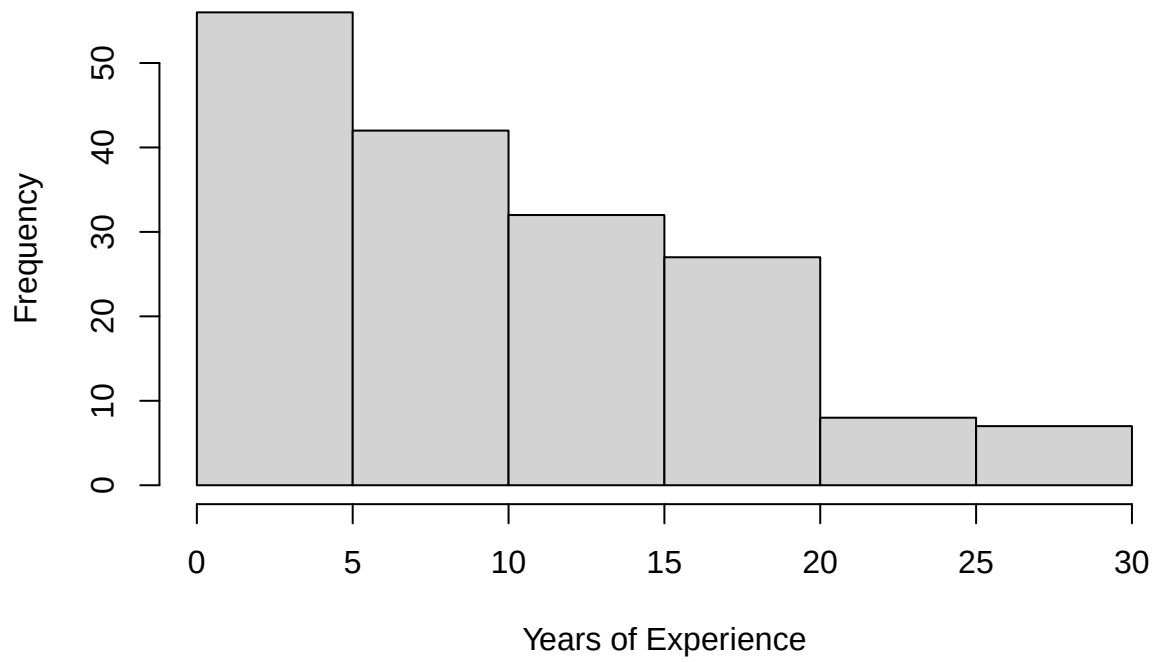
1.2.3 (c) Generate histograms for key numeric variables and describe what you observe about their distributions. Examine correlations between variables.

```
# Create histograms for key numeric variables  
hist(hr$Salary, main = "Histogram of Salary", xlab = "Salary")
```

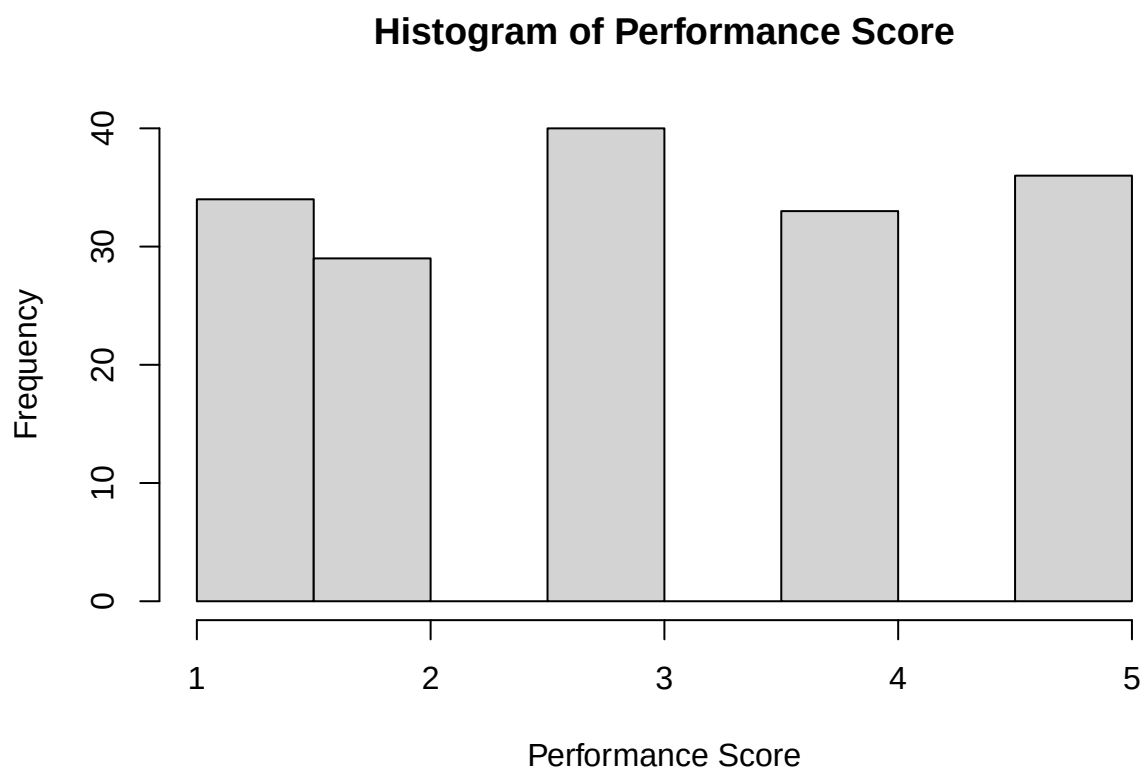


```
hist(hr$YearsOfExperience, main = "Histogram of Years of Experience", xlab = "Years of Experience")
```


Histogram of Years of Experience

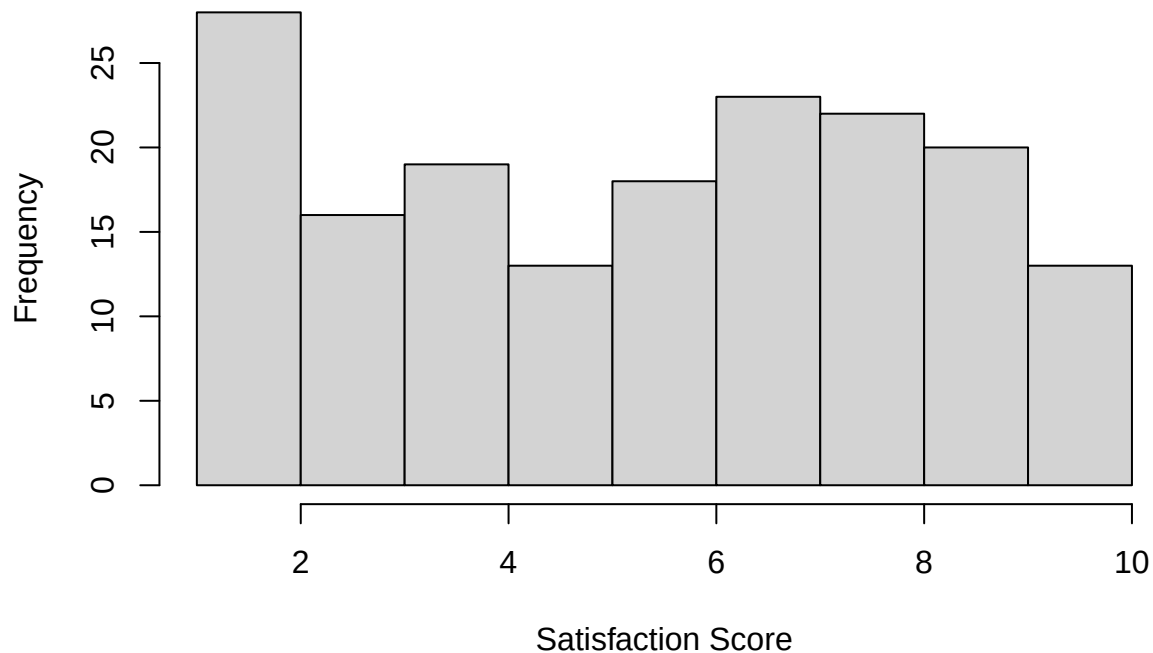


```
hist(hr$PerformanceScore, main = "Histogram of Performance Score", xlab = "Performance Score")
```



```
hist(hr$SatisfactionScore, main = "Histogram of Satisfaction Score", xlab = "Satisfaction Score")
```

Histogram of Satisfaction Score



```
# Examine correlations between all numeric variables
cor_matrix <- cor(numeric_vars)

# Display selected key correlations for easier interpretation
round(cor_matrix[, "Salary"], 2)
```

```
##           Age YearsOfExperience           Salary PerformanceScore
##           0.81           0.82           1.00           0.11
## SatisfactionScore OvertimeHours TrainingHours ProjectsCompleted
##           0.09           -0.07           -0.09           -0.01
##           TeamSize RemoteWorkDays AttendanceRate PromotionCount
##           -0.09           0.00           0.00           0.69
```

```
round(cor_matrix[, "YearsOfExperience"], 2)
```

```
##           Age YearsOfExperience           Salary PerformanceScore
##           0.98           1.00           0.82           0.13
## SatisfactionScore OvertimeHours TrainingHours ProjectsCompleted
##           0.00           -0.11           -0.10           0.01
##           TeamSize RemoteWorkDays AttendanceRate PromotionCount
##           -0.10           0.07           -0.01           0.94
```

```
round(cor_matrix[, "PerformanceScore"], 2)
```

```
##           Age YearsOfExperience           Salary PerformanceScore
##           0.12           0.13           0.11           1.00
## SatisfactionScore      OvertimeHours      TrainingHours ProjectsCompleted
##           -0.12           0.02           -0.12           0.05
##           TeamSize      RemoteWorkDays      AttendanceRate      PromotionCount
##           -0.07           0.06           0.05           0.11
```

```
round(cor_matrix[, "SatisfactionScore"], 2)
```

```
##           Age YearsOfExperience           Salary PerformanceScore
##           0.00           0.00           0.09          -0.12
## SatisfactionScore      OvertimeHours      TrainingHours ProjectsCompleted
##           1.00           0.01           0.03           0.04
##           TeamSize      RemoteWorkDays      AttendanceRate      PromotionCount
##           0.00           0.05           0.08           0.00
```

Histograms also showed that Salary and YearsOfExperience were positively skewed, whereas PerformanceScore and SatisfactionScore were more evenly spread across their scales.

Correlation analysis showed that Salary was strongly positively associated with YearsOfExperience ($r = 0.82$) and Age ($r = 0.81$), and moderately associated with PromotionCount ($r = 0.69$).

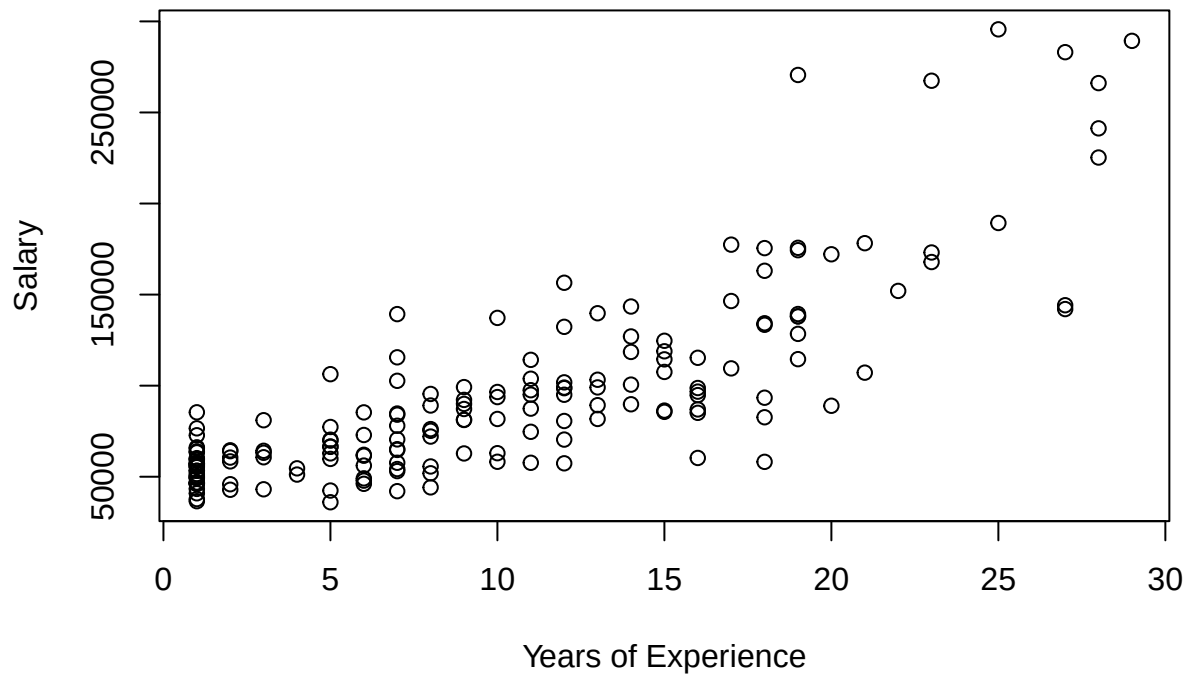
YearsOfExperience was also very strongly correlated with Age ($r = 0.98$) and PromotionCount ($r = 0.94$).

By contrast, PerformanceScore and SatisfactionScore showed only weak relationships with the other numeric variables.

1.2.4 (d) Create appropriate plots (such as box plots, scatter plots, or a correlation heatmap) that reveal interesting patterns or relationships in the data.

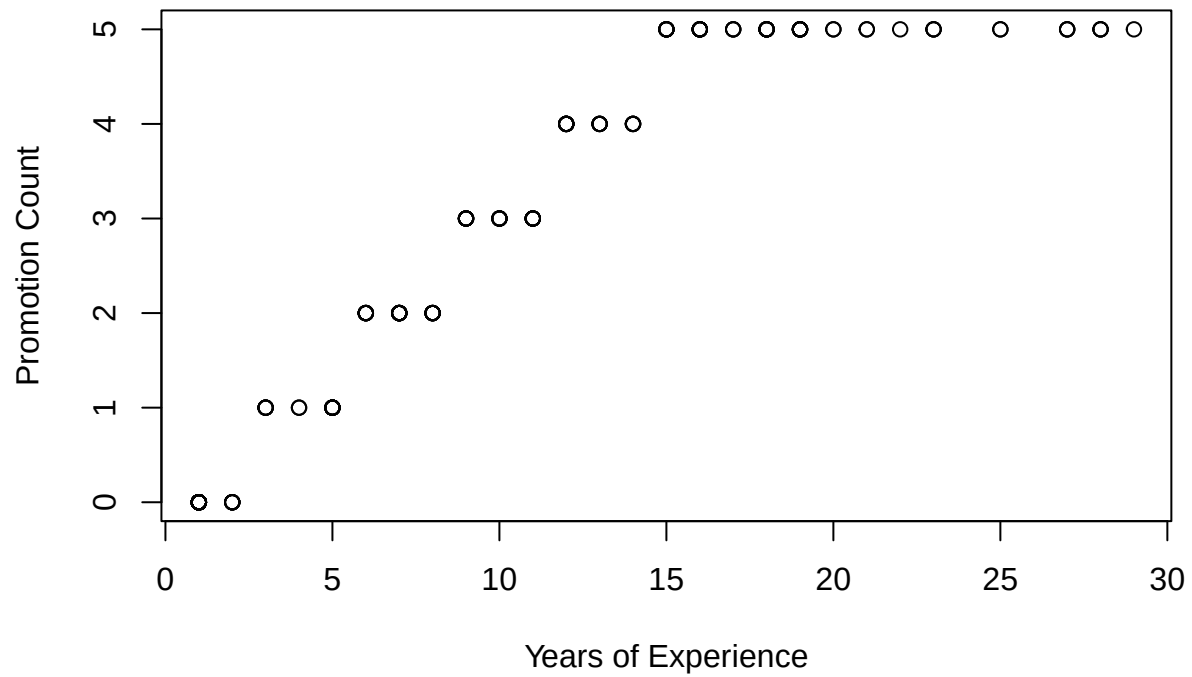
```
# Create a scatter plot for Salary and Years of Experience
plot(hr$YearsOfExperience, hr$Salary,
     main = "Salary vs Years of Experience",
     xlab = "Years of Experience",
     ylab = "Salary")
```

Salary vs Years of Experience



```
# Create a scatter plot for Promotion Count and Years of Experience
plot(hr$YearsOfExperience, hr$PromotionCount,
     main = "Promotion Count vs Years of Experience",
     xlab = "Years of Experience",
     ylab = "Promotion Count")
```

Promotion Count vs Years of Experience



```
# Create a boxplot for Salary by Job Level  
boxplot(Salary ~ JobLevel, data = hr,  
        main = "Salary by Job Level",  
        xlab = "Job Level",  
        ylab = "Salary")
```



Scatter plots confirmed a clear positive relationship between Salary and YearsOfExperience, while boxplots showed salaries increasing across JobLevel from Entry to Executive.

1.2.5 (e) Synthesize your findings into a concise summary of the key characteristics of this workforce.

Overall, the workforce appears to follow a seniority-based compensation pattern: higher experience, age, promotion history, and job level are associated with higher salary, while performance and satisfaction are much less strongly related to pay.

1.2.6 Was the expectation correct?

Yes. Salary was more skewed than the other key numeric variables and was strongly positively associated with YearsOfExperience and JobLevel.

1.3 Error / Warning

The main practical issue was that a single boxplot for all numeric variables was not interpretable because Salary was on a much larger scale. This was resolved by using separate boxplots for selected variables.

The template also mentioned 25 employees, but the actual CSV used for analysis contained 172 observations.

1.4 AI Use

AI was used only to improve the clarity and organisation of the written interpretation after the R outputs had already been produced and checked.

2 Question 2: Comparing Salary Across Job Levels:

2.1 Before analysis

This question examined whether mean salary differed between two job levels. Entry and Mid were selected because they had relatively large and balanced sample sizes. Since salary usually increases with organisational seniority, Mid employees were expected to have a higher mean salary than Entry employees.

2.2 Analysis

2.2.1 (a) Choose two job level groups from the JobLevel variable to compare. Explain your selection and formulate statistical hypotheses.

```
# Count the number of employees in each job level group  
table(hr$JobLevel)
```

```
##  
##      Entry Executive      Mid      Senior  
##        69         8       63         32
```

```
# Create a dataset containing only Entry and Mid employees  
q2_data <- subset(hr, JobLevel == "Entry" | JobLevel == "Mid")
```

```
# Count the number of observations in the selected groups  
table(q2_data$JobLevel)
```

```
##  
## Entry  Mid  
##    69   63
```

```
# Calculate the mean salary for Entry employees  
mean(q2_data$Salary[q2_data$JobLevel == "Entry"])
```

```
## [1] 55839.58
```

```
# Calculate the mean salary for Mid employees  
mean(q2_data$Salary[q2_data$JobLevel == "Mid"])
```

```
## [1] 88383.97
```

```
# Calculate the standard deviation of salary for Entry employees  
sd(q2_data$Salary[q2_data$JobLevel == "Entry"])
```



```
## [1] 10154.83
```

```
# Calculate the standard deviation of salary for Mid employees  
sd(q2_data$Salary[q2_data$JobLevel == "Mid"])
```

```
## [1] 16253.81
```

The descriptive statistics already suggested a large difference. The mean salary was 55,839.58 for Entry and 88,383.97 for Mid, a difference of about 32,544. Salary variation was also larger in the Mid group (SD = 16,253.81) than in the Entry group (SD = 10,154.83).

The null and alternative hypotheses were:

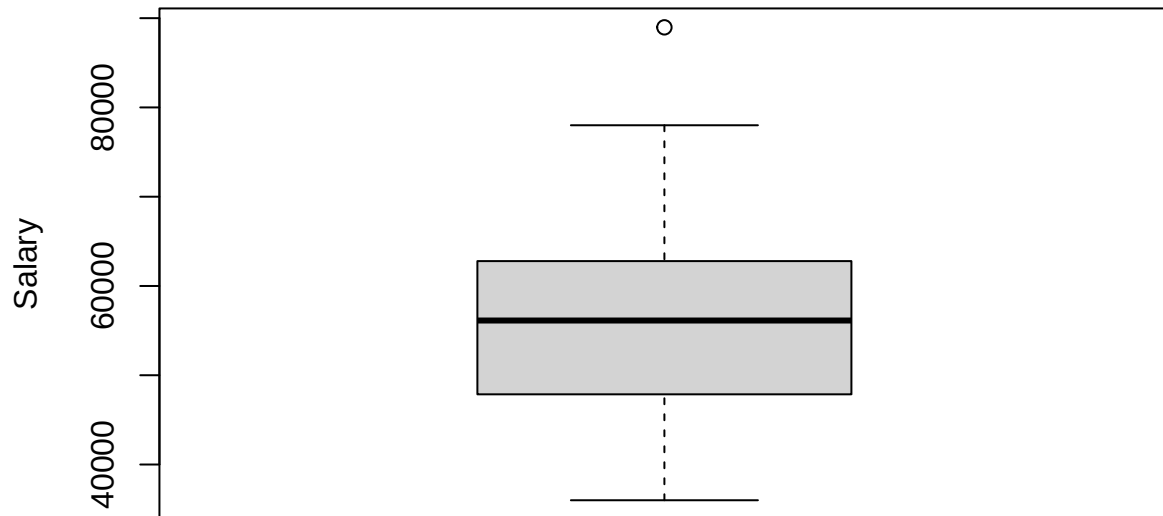
H0: The mean salary is the same for Entry and Mid employees.

H1: The mean salary differs between Entry and Mid employees.

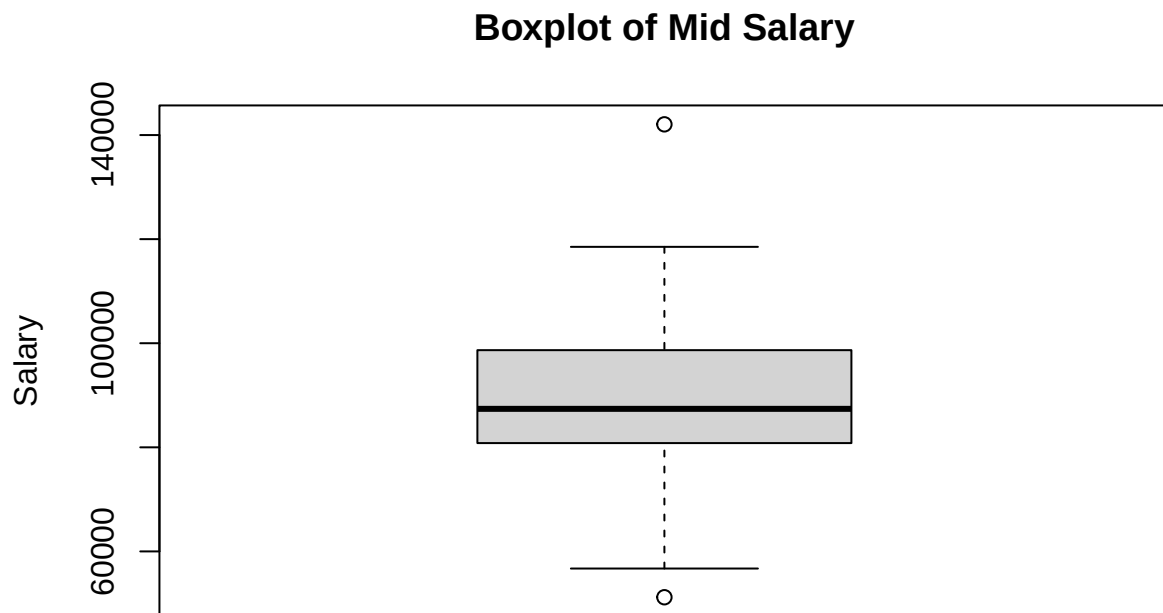
2.2.2 (b) Evaluate whether assumptions for comparing two independent groups are satisfied. Execute the appropriate test and document the test statistic, degrees of freedom, and p-value.

```
# Create separate salary vectors for the Entry and Mid groups  
entry_salary <- q2_data$Salary[q2_data$JobLevel == "Entry"]  
mid_salary <- q2_data$Salary[q2_data$JobLevel == "Mid"]  
  
# Draw boxplots to visually examine the salary distribution in each group  
boxplot(entry_salary, main = "Boxplot of Entry Salary", ylab = "Salary")
```

Boxplot of Entry Salary

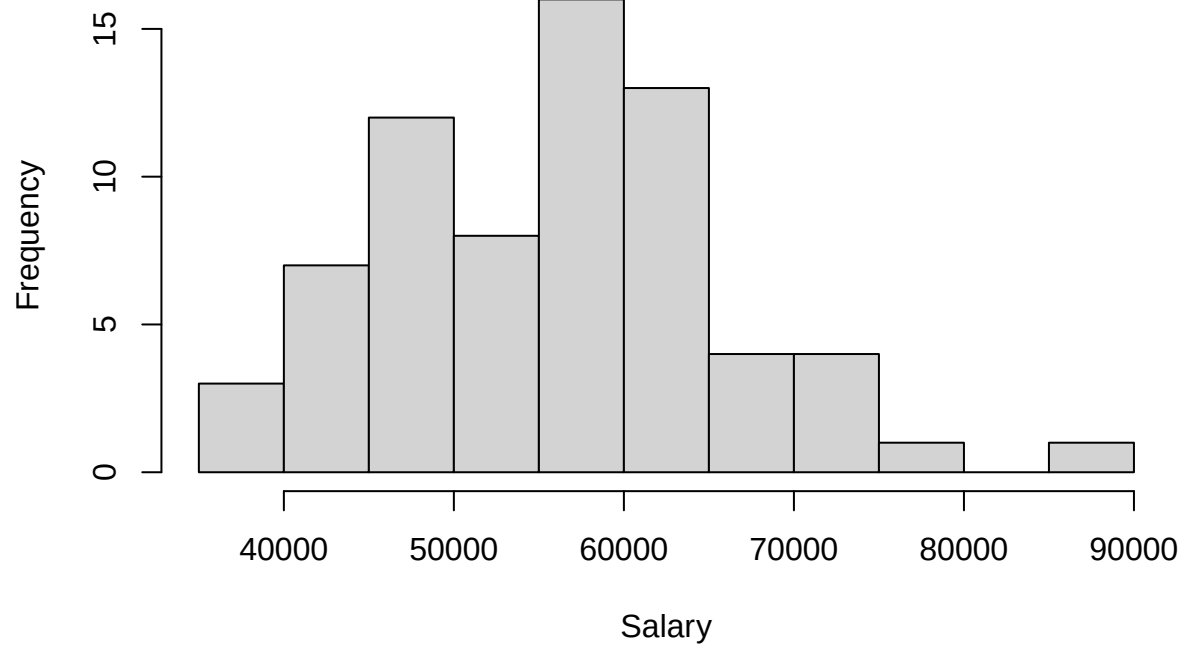


```
boxplot(mid_salary, main = "Boxplot of Mid Salary", ylab = "Salary")
```



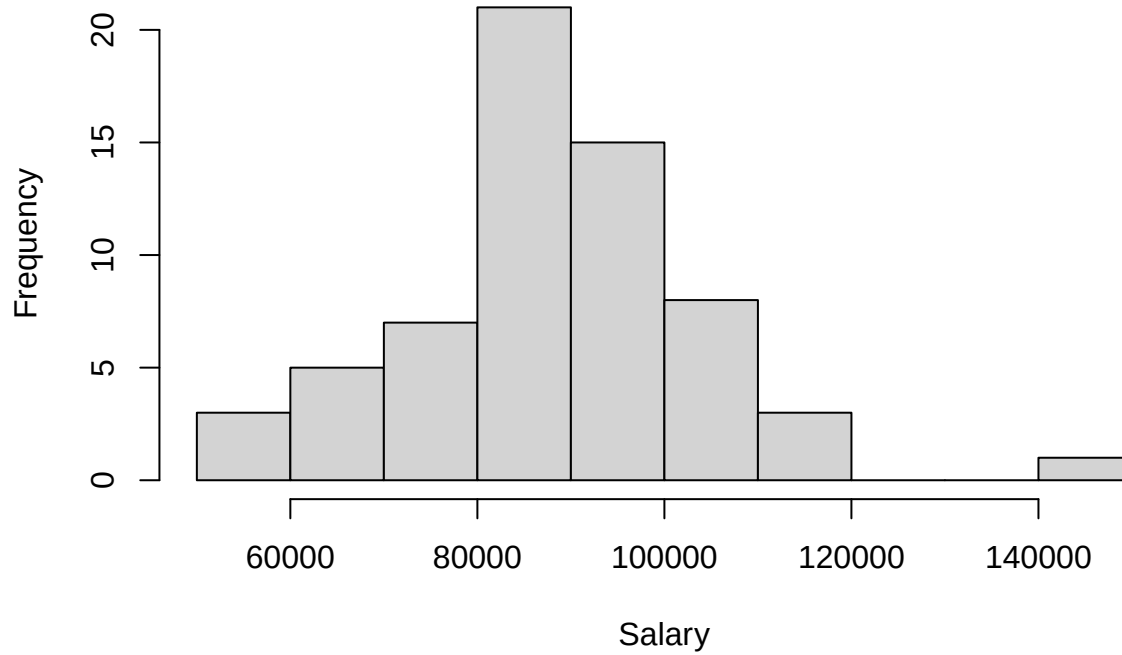
```
# Draw histograms to examine the shape of the salary distribution in each group  
hist(entry_salary, main = "Histogram of Entry Salary", xlab = "Salary")
```

Histogram of Entry Salary



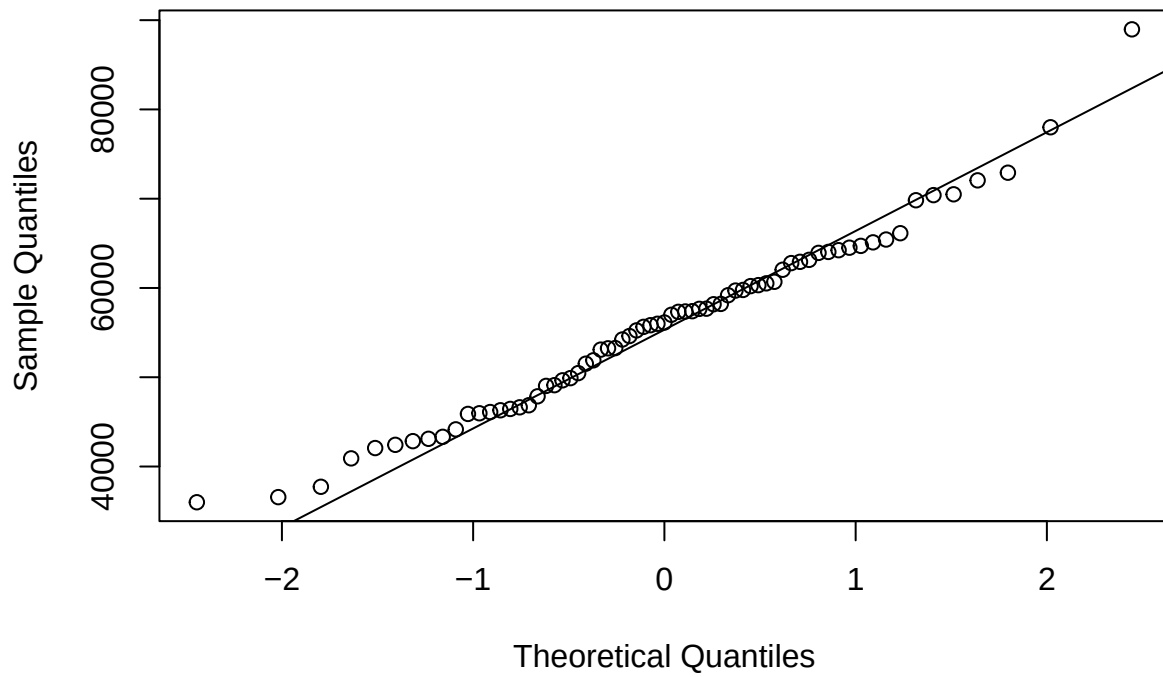
```
hist(mid_salary, main = "Histogram of Mid Salary", xlab = "Salary")
```

Histogram of Mid Salary

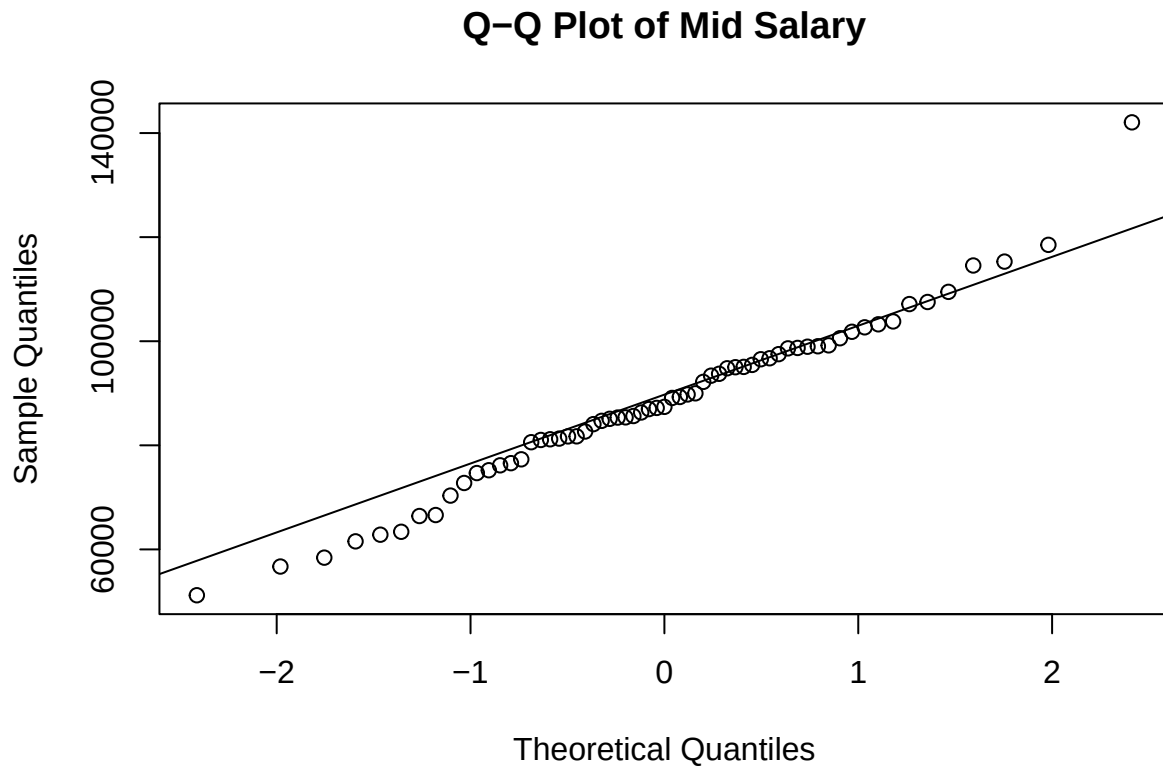


```
# Draw Q-Q plots to assess whether each group is approximately normally distributed  
qqnorm(entry_salary, main = "Q-Q Plot of Entry Salary")  
qqline(entry_salary)
```

Q-Q Plot of Entry Salary



```
qqnorm(mid_salary, main = "Q-Q Plot of Mid Salary")  
qqline(mid_salary)
```



```
# Perform the Shapiro-Wilk normality test for the Entry group  
shapiro.test(entry_salary)
```

```
##  
##  Shapiro-Wilk normality test  
##  
## data:  entry_salary  
## W = 0.98038, p-value = 0.3502
```

```
# Perform the Shapiro-Wilk normality test for the Mid group  
shapiro.test(mid_salary)
```

```
##  
##  Shapiro-Wilk normality test  
##  
## data:  mid_salary  
## W = 0.98018, p-value = 0.4028
```

```
# Test whether the variances of the two groups are equal  
var.test(entry_salary, mid_salary)
```

```
##  
##  F test to compare two variances  
##
```

```
## data:  entry_salary and mid_salary
## F = 0.39033, num df = 68, denom df = 62, p-value = 0.000186
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##  0.2382093 0.6360153
## sample estimates:
## ratio of variances
##           0.3903326

# Perform Welch's two-sample t-test to compare mean salary between the two groups
t.test(entry_salary, mid_salary, var.equal = FALSE)

##
## Welch Two Sample t-test
##
## data:  entry_salary and mid_salary
## t = -13.646, df = 102.23, p-value < 0.00000000000000022
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -37274.78 -27814.00
## sample estimates:
## mean of x mean of y
##  55839.58  88383.97
```

The groups were independent because each employee appeared only once and belonged to one job level. Salary is numeric, so comparing means was appropriate. Histograms, boxplots, and Q-Q plots showed some tail deviation and a few outliers, but the distributions were still reasonably close to normal.

The Shapiro-Wilk tests were non-significant for both Entry ($p = 0.3502$) and Mid ($p = 0.4028$), so there was no strong evidence against normality.

However, the equal variance assumption was not satisfied. The variance test gave $F = 0.3903$ with $p = 0.000186$, indicating unequal variances. Therefore, Welch's two-sample t-test was used.

The test result was $t = -13.646$, $df = 102.23$, $p < 0.001$, providing very strong evidence that the mean salaries differ between the two groups.

2.2.3 (c) Build a 86% confidence interval for the mean difference between groups. Calculate and interpret effect size magnitude.

```
# Construct an 86% confidence interval for the mean difference in salary
t_test_86 <- t.test(entry_salary, mid_salary, var.equal = FALSE, conf.level = 0.86)

# Display the 86% confidence interval and estimated group means
t_test_86

##
## Welch Two Sample t-test
##
## data:  entry_salary and mid_salary
## t = -13.646, df = 102.23, p-value < 0.00000000000000022
## alternative hypothesis: true difference in means is not equal to 0
```



```
## 86 percent confidence interval:
## -36091.63 -28997.15
## sample estimates:
## mean of x mean of y
## 55839.58 88383.97
```

```
# Calculate Glass's delta using the Entry group as the reference group
glass_delta <- (mean(mid_salary) - mean(entry_salary)) / sd(entry_salary)

# Display Glass's delta
glass_delta
```

```
## [1] 3.20482
```

An 86% confidence interval for the mean difference (Entry - Mid) was (-36,091.63, -28,997.15). Because the interval is entirely negative, Entry-level employees earn less on average than Mid-level employees. This suggests that Mid employees earn about 28,997 to 36,092 more on average.

Because the variances were unequal, Glass's delta was used as the effect size, treating Entry as the reference group. The result was 3.20482, indicating a very large effect. Glass's delta was retained because the variance test indicated unequal variances, making a pooled-standard-deviation effect size less suitable in this comparison.

2.2.4 (d) Construct visual representations that facilitate comparison between the two groups.

```
# Create a boxplot to compare salary between Entry and Mid groups
boxplot(Salary ~ JobLevel, data = q2_data,
        main = "Salary by Job Level",
        xlab = "Job Level",
        ylab = "Salary")
```



The boxplot supported this conclusion by showing a clear upward shift in salary from Entry to Mid, with greater spread in the Mid group.

2.2.5 (e) Draw a conclusion at $\alpha = 0.03$. Explain what your results mean for compensation strategy in this organization.

At $\alpha = 0.03$, the null hypothesis is rejected. Salary is clearly differentiated by job level in this organisation, and the difference is both statistically and practically large.

2.2.6 Was the expectation correct?

Yes. Mid-level employees had a substantially higher mean salary than Entry-level employees, consistent with the expected compensation structure.

2.3 Error / Warning

The main issue was that the equal variance assumption was not met, so the standard pooled two-sample t-test was not appropriate. This was resolved by using Welch's t-test. Another point requiring care was that the confidence interval had to be set to 86%, not the more common 95%.

2.4 AI Use

AI was used to help review possible effect size measures after unequal variances were identified. It was helpful for considering Glass's delta, but the final method was chosen only after checking the variance results.

3 Question 3: Analyzing Categorical Variable Relationships:

3.1 Before analysis

This question tested whether Gender and JobLevel were associated. A strong relationship was not expected, but a formal test was required.

3.2 Analysis

3.2.1 (a) Construct a contingency table examining the relationship between Gender and JobLevel (or two other relevant categorical variables). State hypotheses for testing independence.

```
# Create a contingency table for Gender and JobLevel
gender_joblevel_table <- table(hr$Gender, hr$JobLevel)
```

```
# Display the contingency table
gender_joblevel_table
```

```
##
##      Entry Executive Mid Senior
## Female      28         4   34    17
## Male       37         4   27    15
## Other       4         0    2     0
```

A contingency table was constructed for Gender and JobLevel. The hypotheses were:

H0: Gender and JobLevel are independent.

H1: Gender and JobLevel are not independent.

3.2.2 (b) Compute expected frequencies under independence and execute the appropriate statistical test. Report the test statistic, degrees of freedom, and p-value.

```
set.seed(123)
```

```
# Perform chi-square test of independence with simulated p-value
chi_test <- chisq.test(gender_joblevel_table, simulate.p.value = TRUE, B = 10000)
```

```
# Display full test result
chi_test
```

```
##
## Pearson's Chi-squared test with simulated p-value (based on 10000
## replicates)
##
## data:  gender_joblevel_table
## X-squared = 4.7084, df = NA, p-value = 0.5819
```

```
# Display expected frequencies under independence
chi_test$expected
```

```
##
##           Entry Executive      Mid      Senior
##   Female 33.296512 3.8604651 30.401163 15.441860
##   Male   33.296512 3.8604651 30.401163 15.441860
##   Other   2.406977 0.2790698  2.197674  1.116279
```

```
# Manually calculate degrees of freedom
df_q3 <- (nrow(gender_joblevel_table) - 1) * (ncol(gender_joblevel_table) - 1)
df_q3
```

```
## [1] 6
```

Expected frequencies were then calculated under independence.

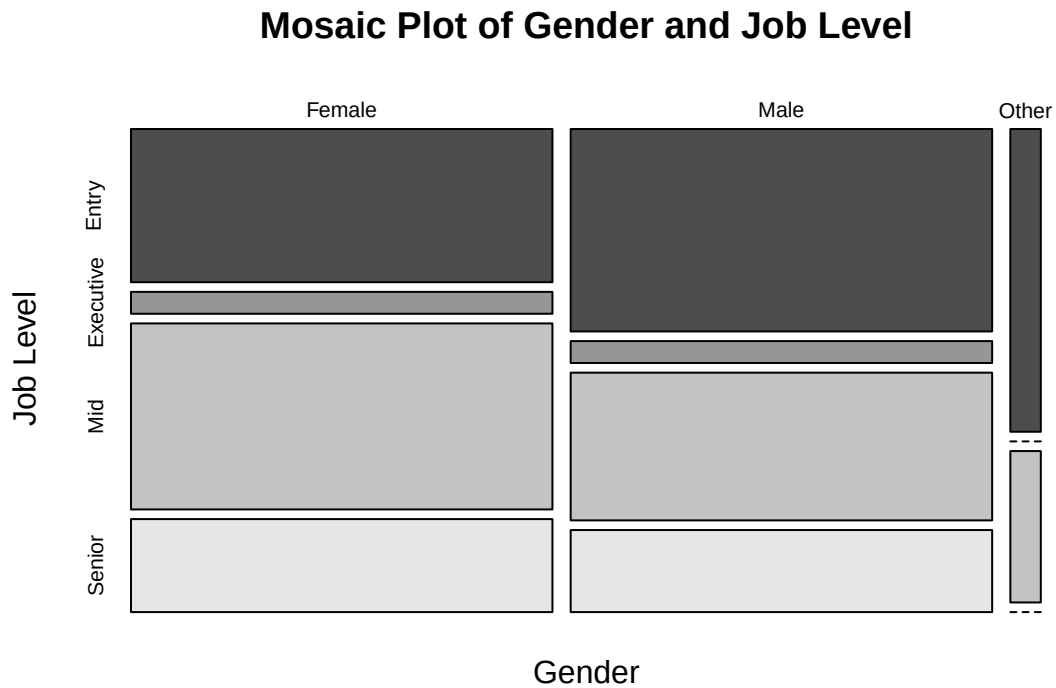
Some expected counts were small, especially in cells involving the Other category, with several below 5. Because this weakens the reliability of the standard chi-square approximation, a chi-square test with simulated p-value (10,000 replicates) was used instead.

The result was chi-square = 4.7084, df = 6, p = 0.5819.

Since the p-value is much larger than $\alpha = 0.03$, the null hypothesis is not rejected.

3.2.3 (c) Develop a clear visualization (mosaic plot or stacked bar chart) that illustrates the categorical relationship.

```
# Create a mosaic plot for Gender and JobLevel
mosaicplot(gender_joblevel_table,
  main = "Mosaic Plot of Gender and Job Level",
  xlab = "Gender",
  ylab = "Job Level",
  color = TRUE)
```



A mosaic plot showed that the Female and Male categories had broadly similar distributions across the four job levels. Entry and Mid were the largest categories in both groups, while Executive was much smaller.

The Other category had very few observations and should therefore be interpreted cautiously.

3.2.4 (d) Render a decision at $\alpha = 0.03$. Discuss what your analysis reveals about organizational diversity and structure.

Overall, there is no statistically significant evidence that Gender and JobLevel are associated in this dataset.

From an organisational diversity perspective, the distribution of employees across job levels appears broadly similar across the gender categories represented in the sample.

3.2.5 Was the expectation correct?

Yes. The test did not detect a significant association between Gender and JobLevel.

3.3 Error / Warning

Some expected frequencies were small, especially for the Other category. This was handled by using a simulated p-value rather than relying only on the standard asymptotic chi-square result.

3.4 AI Use

AI was used mainly to improve the wording of the conclusion so that the non-significant result was not overstated and the limitation from small expected counts was acknowledged clearly.

4 Question 4: Simple Linear Regression: Predicting Salary from YearsOfExperience:

4.1 Before analysis

This question examined whether YearsOfExperience could predict Salary using simple linear regression. Based on the EDA, a positive relationship was expected.

4.2 Analysis

4.2.1 (a) Fit the simple linear regression model in R using `lm()`. Report the estimated regression equation: $\hat{y} = b_0 + b_1x$.

```
# Fit the simple linear regression model
model_salary_exp <- lm(Salary ~ YearsOfExperience, data = hr)

# Display the model summary
summary(model_salary_exp)

##
## Call:
## lm(formula = Salary ~ YearsOfExperience, data = hr)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -82620 -15549   -129   13418 124008
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    36921.8     3882.5    9.51 <0.0000000000000002 ***
## YearsOfExperience    5770.9       312.8   18.45 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 30430 on 170 degrees of freedom
## Multiple R-squared:  0.667, Adjusted R-squared:  0.665
## F-statistic: 340.4 on 1 and 170 DF, p-value: < 0.00000000000000022

# Extract coefficients
coefs <- coef(model_salary_exp)
b0 <- coefs[1]
b1 <- coefs[2]
```

The fitted regression equation was:

Salary = 36921.8 + 5770.9 * YearsOfExperience


```
par(mfrow = c(1, 1))
```

To test whether YearsOfExperience is a significant predictor, the hypotheses were:

H0: $b_1 = 0$ H1: $b_1 \neq 0$

The regression output gave $t = 18.45$ for the slope with $p < 2e-16$, so the null hypothesis is rejected at $\alpha = 0.05$. There is very strong evidence of a positive linear relationship between YearsOfExperience and Salary.

Diagnostic plots suggested only minor deviations from ideal assumptions, so the model remained a useful summary of the relationship.

4.2.4 (d) Report and interpret R-squared. What proportion of variation in Salary is explained by YearsOfExperience?

```
# Display the R-squared value
summary(model_salary_exp)$r.squared
```

```
## [1] 0.6669587
```

The R-squared value was 0.6670, meaning that about 66.7% of the variation in Salary is explained by YearsOfExperience alone. This indicates that experience is an important predictor of pay, although other factors also likely explain the remaining variation.

4.2.5 (e) Use the model to predict the salary for an employee with 5 years of experience, including appropriate interpretation. Discuss practical implications for compensation strategy.

```
# Predict salary for 5 years of experience
new_data <- data.frame(YearsOfExperience = 5)
pred_salary <- predict(model_salary_exp, newdata = new_data)
pred_salary
```

```
##          1
## 65776.13
```

For an employee with 5 years of experience, the predicted salary is 65,776.13. This is the average salary expected from the fitted model for someone with that level of experience.

From an HR compensation perspective, the model suggests that experience is a major factor in salary determination, but not the only one.

4.2.6 Was the expectation correct?

Yes. YearsOfExperience had a positive and statistically significant relationship with Salary, as expected.

4.3 Error / Warning

The main issue was ensuring that the written equation, slope test, R-squared value, and prediction all matched the final regression output. This was checked directly against the model summary before finalising the interpretation.

4.4 AI Use

AI was used only to improve the wording of the interpretation so that the regression results were explained clearly in the context of compensation strategy.

5 Question 5: Comparing Performance Score Across Job Levels:

5.1 Before analysis

This question tested whether mean PerformanceScore differed across the four job levels. Large differences were not expected because performance may be assessed relative to the expectations of each role.

5.2 Analysis

5.2.1 (a) Create a box plot showing Performance Score distributions across all Job Levels.

```
# Convert JobLevel to factor in the desired order
hr$JobLevel <- factor(hr$JobLevel, levels = c("Entry", "Mid", "Senior", "Executive"))

# Create boxplot of PerformanceScore by JobLevel
boxplot(PerformanceScore ~ JobLevel, data = hr,
        main = "Performance Score by Job Level",
        xlab = "Job Level",
        ylab = "Performance Score")
```



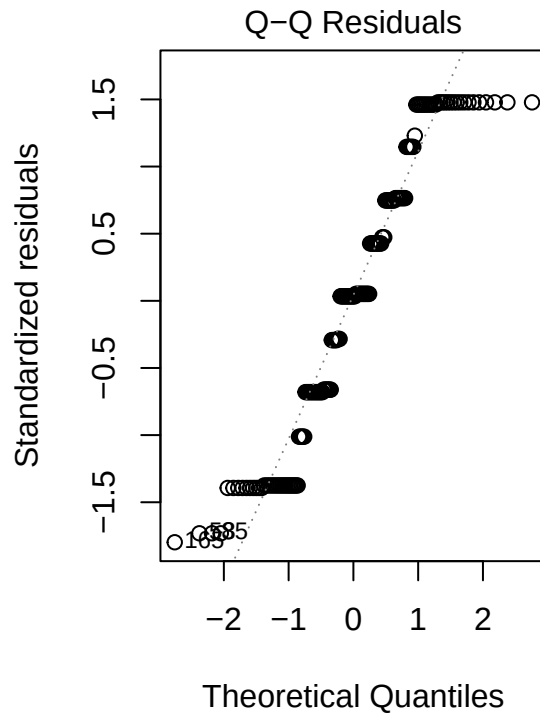
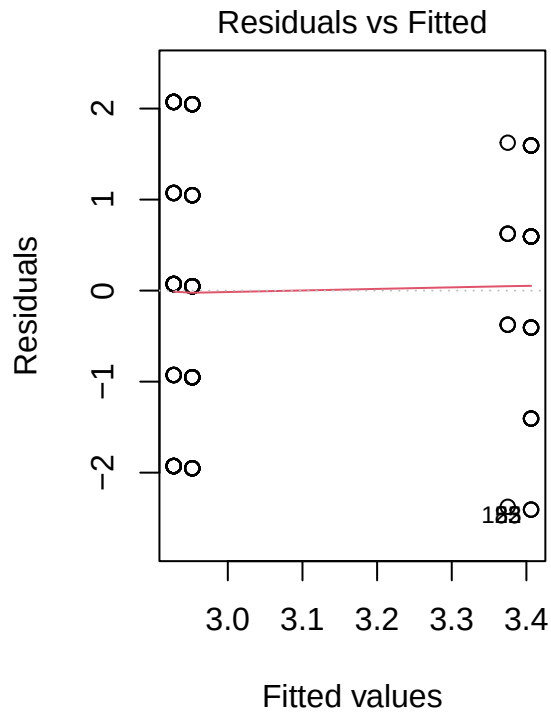
A boxplot showed broadly similar PerformanceScore distributions across Entry, Mid, Senior, and Executive employees.

Senior employees appeared to have a slightly higher median, but the interquartile ranges overlapped considerably.

5.2.2 (b) Perform the hypothesis testing to determine whether the Performance Score are the same across all Job levels.

```
# Fit one-way ANOVA model
anova_perf <- aov(PerformanceScore ~ JobLevel, data = hr)

# Check ANOVA assumptions
par(mfrow = c(1, 2))
plot(anova_perf, which = 1)
plot(anova_perf, which = 2)
```



```
par(mfrow = c(1, 1))

# Test homogeneity of variance
bartlett.test(PerformanceScore ~ JobLevel, data = hr)

##
## Bartlett test of homogeneity of variances
##
## data: PerformanceScore by JobLevel
## Bartlett's K-squared = 2.8826, df = 3, p-value = 0.4101
```

```
# Test normality of ANOVA residuals
shapiro.test(residuals(anova_perf))
```

```
##
## Shapiro-Wilk normality test
##
## data: residuals(anova_perf)
## W = 0.9237, p-value = 0.00000007424
```

```
# Display ANOVA results
summary(anova_perf)
```

```
##           Df Sum Sq Mean Sq F value Pr(>F)
```

```
## JobLevel      3      6.5    2.180    1.093    0.354
## Residuals    168    335.1    1.995
```

The one-way ANOVA hypotheses were:

H0: Mean PerformanceScore is the same across all job levels.

H1: At least one job level has a different mean PerformanceScore.

Before interpreting the ANOVA, the assumptions were checked. Bartlett's test gave $p = 0.4101$, so there was no strong evidence of unequal variances.

The Shapiro-Wilk test on the residuals suggested non-normality, which is not surprising because PerformanceScore is a discrete 1-to-5 variable. Although residual normality was imperfect, ANOVA was retained because the group sizes were moderate and the method is reasonably robust to mild deviations from normality. The results should still be interpreted with some caution.

The ANOVA result was $F = 1.093$, $p = 0.354$. Since the p-value is greater than 0.05, the null hypothesis is not rejected. There is no statistically significant evidence that the mean PerformanceScore differs across job levels.

5.2.3 (c) If the overall test is significant, conduct Tukey's HSD post-hoc test. Report which specific job level pairs differ significantly.

Because the overall ANOVA was not significant, Tukey's HSD was not required.

5.2.4 (d) Interpret your findings: What does this reveal about performance across job levels in the organization?

Overall, the analysis suggests that average performance is broadly similar across organisational levels in this dataset.

This may indicate that performance is assessed relative to each role's expectations rather than increasing automatically with seniority.

5.2.5 Was the expectation correct?

Yes. The results were consistent with the expectation that performance scores would not differ greatly across job levels.

5.3 AI Use

AI was used only as a second reader for the interpretation, mainly to ensure that the conclusion stayed cautious and did not overstate the small visual differences seen in the boxplot.